5 A trolley runs down a slope with a constant acceleration *a*. The mass of the trolley is now doubled, and the trolley is allowed to run down the same slope. In both cases effects of friction and air resistance are negligible.

A The accelerating force is the same.

B The acceleration is $\frac{1}{2}a$.

C The acceleration is *a*.

D The acceleration is 2*a*.

Ans: C

For a trolley on a frictionless slope, the only force causing acceleration is the component of gravity along the slope: $F = mg\sin\theta$

The acceleration is: $a = \frac{F}{m} = \frac{mg\sin\theta}{m} = g\sin\theta$

When Mass is Doubled

The new accelerating force becomes: $F' = 2mg\sin\theta$

So, the accelerating force is doubled — meaning option A is incorrect.

The new acceleration is:

$$a' = \frac{F'}{2m} = \frac{2mg\sin\theta}{2m} = g\sin\theta = a$$

The mass cancels out completely, just like free fall.

6 A force of 54 N pushes two touching blocks of mass 6.0 kg and 2.0 kg along a flat surface.